Historical changes and pipeline resilience

Historical changes

I would preserve successive source versions instead of overwriting them. The ingestion-added _ingested_at timestamp records when each version first became available and must remain unchanged during replay. SQL derives availability intervals and selects the version known at each calendar month-end. Later versions therefore do not replace earlier month-end profile, contract or invoice states.

Inventory and geography use the latest complete snapshot dated on or before month-end. Ingestion must ensure that eligible snapshots were available by that cutoff. For a mid-month move, the mart reflects the service location recorded in those snapshots. Monthly geography snapshots cannot establish the exact move date or reconstruct unobserved intermediate states. Where effective dates and change events are available, a Type 2 slowly changing dimension can additionally preserve business-valid intervals. Availability timestamps and business-effective dates must remain distinct; missing past versions cannot be recreated.

Idempotency and failure recovery

At ingestion, stable delivery identifiers and a load ledger must prevent duplicate batch loading. Replayed versions retain their original availability timestamps, and incomplete snapshots must remain ineligible.

In the proposed design, Airflow verifies expected deliveries before passing sources_ready to SQL and prevents concurrent publications over overlapping month ranges. Transformations build temporary candidate tables; quality checks must pass before publication.

The final table is partitioned by billing_month and clustered by customer_id. Publication replaces only requested months using DELETE and INSERT within one transaction. COMMIT validates both operations together. Outside session mode, an unhandled script error automatically rolls back the open transaction. Other months remain unchanged. With identical validated inputs, rerunning the same period replaces its rows rather than appending duplicates.